

LAB ASSIGNMENT – 3

Question 1

Data Instructions

In this lab, synthetic labelled datasets are considered for a multi-class classification problem. The number of classes is fixed to $K = 3$. Let N denote the total number of samples in a dataset, where the dataset size is varied as:

$$N \in \{50, 100, 1000\}.$$

Each data point belongs to exactly one of the three classes. Class labels are generated independently according to a predefined categorical distribution characterized by known class prior probabilities.

Two different class-prior configurations are used in this study:

- (a) **Moderately imbalanced class distribution**, defined by the prior probability vector:

$$\boldsymbol{\pi} = [0.7, 0.2, 0.1].$$

- (b) **Highly imbalanced class distribution**, defined by the prior probability vector:

$$\boldsymbol{\pi} = [0.9, 0.08, 0.02].$$

For each dataset, generate class labels independently according to the specified priors. Encode the class labels using a 1-of- K (one-hot) representation.

Tasks

Using Python, perform the following steps:

1. For each dataset size N , generate a labelled dataset using the given class priors.
2. Estimate the class prior probabilities using the maximum likelihood estimation (MLE) formula:

$$\hat{\pi}_k = \frac{n_k}{N},$$

where n_k is the number of samples belonging to class k .

3. Repeat the estimation for both the moderately imbalanced and highly imbalanced class distributions.
4. Plot the estimated class priors as a function of dataset size and compare them with the true class priors.
5. Analyze how increasing dataset size affects the convergence of the estimated priors, with particular attention to the minority classes.

Question 2

Data Instructions

You must generate synthetic two-dimensional datasets using Python with two classes C_1 and C_2 . Each class contains N samples, where:

$$N \in \{20, 50\},$$

and the feature space is $\mathbb{R}^2$.

You will generate two datasets:

- (a) **Linearly separable dataset:** The convex hulls of the two classes do not intersect.
- (b) **Non-linearly separable dataset:** The convex hulls of the two classes intersect.

Tasks

Using Python, perform the following steps:

1. Generate the two datasets according to the specifications above.
2. Train a binary logistic regression classifier on each dataset.
3. Plot the data points along with the learned logistic regression decision boundary.
4. Evaluate the classification performance using training accuracy.
5. Compute and visualize the convex hull of each class.
6. Verify whether the convex hulls intersect in each case.
7. Analyze the relationship between convex hull intersection, logistic regression performance, and the existence of a linear decision boundary.